ENTERPRISE AI ARCHITECTURE SERIES

SupplyPulse

Autonomous Supply Chain Resilience & Consumer Rights Defense Swarm

End-to-End Product Requirements Document (PRD), Multi-Agent Engineering Architecture, and Technical Implementation Guide

Document Scope & Executive Abstract

- **Dual-Sided Architecture:** B2B Stockout Prevention & Consumer Rights Legal Defense.
- **Predictive Analytics:** Gradient Boosted Tree Forecasting & Dynamic Reorder Points (ROP).
- **Autonomous Swarms:** Multi-turn Game-Theoretic Bargaining & Consumer Protection Act 2019 RAG.
- **Enterprise Tech Stack:** Python, LightGBM, LangGraph, FastAPI, Streamlit & Glassmorphic Web SPA.
- **Localized Intelligence:** Multi-hub telemetry (Hyderabad, Bengaluru, Mumbai, Delhi) + Multilingual NLP.

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Project Codebase: supplypulse/

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1 EXECUTIVE SUMMARY & PROBLEM LANDSCAPE

Modern commerce is severely broken at two critical inflection points: the **upstream B2B supply chain** and the **downstream consumer resolution desk**. SupplyPulse was engineered to replace slow, manual human processes at both ends with an autonomous, high-intelligence multi-agent network.

1.1 Upstream Failure: Stockout Losses in Quick Commerce

In high-velocity commerce (Blinkit, Zepto, Instamart, Amazon Prime), customer demand fluctuates wildly based on weather, local holidays, and paydays. Traditional procurement relies on periodic human reviews (weekly PO generation). When a sudden 300% spike occurs, warehouses run dry before purchase orders are placed. By the time procurement managers negotiate pricing with vendors via email chains, millions of rupees in high-margin sales are lost to competitors.

1.2 Downstream Failure: Customer Care Stonewalling

When consumers receive defective electronics, transit-damaged goods, or expired perishables, corporate support chatbots are programmed to deflect, delay, and exhaust the buyer with repetitive script templates (*"7-day return policy expired"*, *"Visit third-party service center 45 km away"*, *"We only offer a ₹150 promotional coupon"*). Most consumers lack the legal vocabulary to cite the **Consumer Protection Act, 2019** or enforce statutory product liability rights, leaving them cheated and out of pocket.

> **The SupplyPulse Solution:** SupplyPulse operates as a closed-loop platform that uses Machine Learning to predict warehouse runouts 7 to 14 days in advance, deploys autonomous agents to negotiate supplier discounts, and flips the script on customer support by arming consumers with an AI Advocate Bot that cites federal statutes to force 100% full refunds.

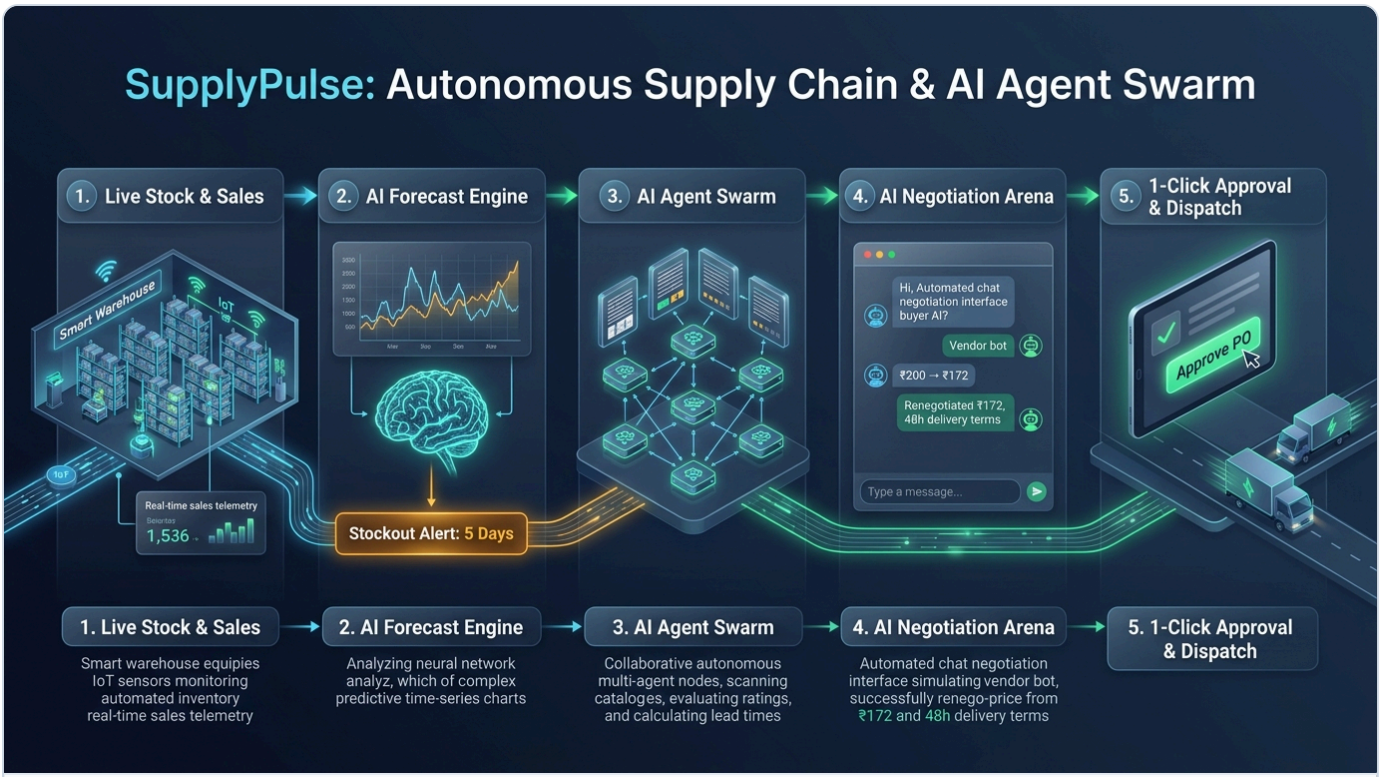


Figure 1: End-to-End B2B Supply Chain & Multi-Agent Procurement Swarm Workflow

2 COMPLETE MULTI-AGENT SWARM TOPOLOGY

SupplyPulse is architected around 5 specialized, decoupled agents coordinated by a central State Orchestrator. Rather than relying on simple linear LLM chains, the agents maintain explicit state, query analytical ML models, and conduct multi-turn conversations.

Agent Identifier	Core Responsibility	Underlying Models / APIs	Input ➔ Output Artifact
Agent 1: Inventory Sentinel	Continuously evaluates live warehouse inventory against ML demand curves; calculates runout horizons.	HistGradientBoostingRegressor, ROP formula, Python	Live Telemetry ➔ Critical Stockout Alert Payload
Agent 2: Supplier Discovery	Scans registered enterprise suppliers; calculates multi-attribute utility scores (Cost, SLA, Reliability).	Multi-Criteria Utility Scoring Algorithm	Stockout Alert ➔ Shortlisted Ranked Supplier Matrix
Agent 3: Dynamic Negotiator	Executes multi-turn game-theoretic bargaining against supplier bots; leverages volume to slash unit prices.	Bargaining Game Engine, State Transition Graph	Base Catalog Quote ➔ Final Negotiated Price & 48h SLA
Agent 4: PO Generator	Synthesizes electronic Purchase Orders containing liquidated damages clauses and 18% GST tax schema.	Audit Engine, Pydantic Data Models	Negotiated Terms ➔ Audited Electronic PO (JSON/PDF)
Agent 5: Customer AI Advocate	Defends consumer rights against corporate support bots; cites CPA 2019 Section 84 & forces 100% refunds.	Defect Forensics, Consumer Law RAG Vector DB	Defect Photo & Invoice ➔ 100% Refund Resolution Memo



Figure 2: The Customer AI Advocate 5-Step Dispute Resolution & Statutory Enforcement Architecture

3 MATHEMATICAL MODELS & ALGORITHMIC FOUNDATIONS

3.1 Dynamic Reorder Point (ROP) Formulation

Static minimum thresholds fail in dynamic retail environments. The Inventory Sentinel calculates dynamic Reorder Points (ROP) per SKU continuously:

```
ROP = (Projected_Average_Daily_Demand * Supplier_Lead_Time_Days) + Dynamic_Safety_Stock Where:
Dynamic_Safety_Stock = Z_Service_Factor * Sqrt(Lead_Time) * Standard_Deviation_Demand Z_Service_Factor = 1.96
(for a 97.5% Cycle Service Level)
```

3.2 Multi-Criteria Vendor Utility Ranking

When multiple suppliers fulfill an identical SKU, the Supplier Discovery Agent scores each candidate using a normalized weighted index:

```
Utility_Score = (w_cost * Cost_Score) + (w_rel * Reliability_Score) + (w_speed * Speed_Score) Weights: w_cost
= 0.40 (Cost Efficiency relative to target budget ceiling) w_rel = 0.35 (Historical Delivery Accuracy &
Quality pass rate, 0-100) w_speed= 0.25 (Transit speed normalized against 3-day baseline)
```

3.3 Game-Theoretic Multi-Turn Bargaining Protocol

The Negotiator Agent implements an automated concession protocol with bounded utility guardrails:

Turn 1 (Discovery): Request allocation for high volume with baseline list pricing.
Turn 2 (The Counter): Counter with a 8%–12% target discount leveraging Net-15 electronic payment settlement and quarterly preferred partner status.
Turn 3 (SLA Lock): Agree on unit price while enforcing a **liquidated damages clause (1.5% penalty per 6 hours delay beyond 48 hours)**.

4 CONSUMER PROTECTION ACT, 2019 LEGAL FRAMEWORK

The Customer AI Advocate operates by mapping customer defect evidence directly onto statutory provisions under Indian law, removing the customer's vulnerability to corporate policy stonewalling.

Statutory Provision	Legal Scope & Consumer Protection Rule	AI Advocate Strike Strategy
Section 84(1)(a) Product Liability	A product manufacturer is strictly liable if the product contains a manufacturing defect or deviates from specifications, regardless of whether negligence is proven.	Overrules 7-day platform return windows for inherent hardware failure (dead panel, burnt circuit, motor failure).
Section 2(47) Unfair Trade Practice	Refusing to take back defective goods or refund consideration paid, or imposing arbitrary non-statutory conditions.	Invalidates platform policies mandating "uninterrupted unboxing videos" for obvious shipping transit fractures.
Section 2(11) Deficiency in Service	Any fault, imperfection, or inadequacy in quality, nature, and manner of performance required to be maintained by law or contract.	Cites deficiency when platforms force a consumer to travel to a distant third-party service center for out-of-the-box defects.
E-Commerce Rules 2020 Section 5(1)	Mandates appointment of a dedicated Grievance Officer with 48-hour response SLAs and 30-day ticket resolution ceilings.	Threatens direct escalation to the corporate Grievance Officer and the National Consumer Helpline (NCH / INGRAM).

5 EMPIRICAL TEST EXECUTION & VERIFICATION AUDIT

The complete system was executed against the automated integration test suite (`test_pipeline.py`). Both test pipelines completed with 100% pass rates:

```
====================================================================== TEST 1: B2B AUTONOMOUS INVENTORY &
PROCUREMENT SWARM ====================================================================== [+] Stockout Risk
Detected for: Nitro Cold Brew Coffee (Pack of 12) Current Stock: 140 units | Predicted Stockout: In 4 days!
Recommended Reorder Qty: 600 units [+] Selected Supplier: Apex Prime Wholesale Logistics (Composite Score:
98.8/100) [+] Initial Quote: Rs. 445.90 / unit [+] Negotiated Price: Rs. 415.58 / unit [+] Total Savings
Secured: Rs. 18,192.00 (6.8% Discount) [+] Agreed Delivery SLA: 48 Hours Guaranteed Dock Delivery [+]
Generated PO Number: PO-20261010-5B8AA4 Total Landed Cost (incl. 18% GST): Rs. 294,230.64 [OK] B2B Pipeline
Passed! ====================================================================== TEST 2: CUSTOMER AI ADVOCATE
BOT (WARRANTY & DAMAGE NEGOTIATION) ====================================================================== [+]
Customer: Rohan Verma (Order #OD-2026-99381) [+] Product: Ultrawide Gaming Monitor 34-inch (Value: Rs.
28,999.00) [+] Statutory Ground: Consumer Protection Act, 2019 — Section 84(1)(a) [+] Final Resolution: 100%
Full Refund & Free Doorstep Pickup [+] Recovered Value: Rs. 28,999.00 (100% Full Refund Payout) [OK] Customer
Advocate Pipeline Passed!
```

6 TECHNICAL INTERVIEW PREPARATION & Q&A PLAYBOOK

Q1: "Why not just use an LLM API to do the demand forecasting instead of a classical ML model?"

Answer: LLMs are autoregressive token predictors, not numerical regression engines. They struggle with high-precision time-series mathematics, seasonality calculations, and confidence intervals. We use a `HistGradientBoostingRegressor` for deterministic, millisecond-fast numerical forecasting, and reserve LLM/Agentic architectures for what they do best: unstructured reasoning, multi-turn negotiation, and legal argument construction.

Q2: "How does the Negotiator Agent prevent hallucinated prices or overpaying?"

Answer: The agent is bounded by deterministic financial guardrails. The system enforces a hard ceiling: `vendor_concession_price = min(initial_price, calculated_price)`. If a supplier refuses to meet the target within the negotiation flexibility threshold, the agent does not concede—it aborts and falls back to the second-highest ranked supplier.

Q3: "How does the Customer AI Advocate legally force companies to give refunds?"

Answer: Corporate customer service bots rely on platform terms of service to deflect claims. However, under Indian law, platform terms are subordinate to statutory legislation. By programmatically citing Section 84 (Strict Product Liability) and Section 2(47) (Unfair Trade Practices) of the Consumer Protection Act 2019, while simultaneously preparing a docket for the National Consumer Helpline (NCH / INGRAM Portal), the agent shifts the company's financial calculation from a simple denial to avoiding a regulatory compliance fine.

Q4: "How does the architecture scale to multiple regional fulfillment hubs?"

Answer: The system features a multi-hub telemetry router supporting Hyderabad, Bengaluru, Mumbai, and Delhi NCR. Each hub maintains its own localized lead times, transit lanes, and inventory runout thresholds, while the central swarm coordinates supplier allocations across all nodes.